MILITARY SPECIFICATION

BEARINGS, SLEEVE, PLAIN AND FLANGED, SELF-LUBRICATING, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers plain and flanged sleeve bearings which are self-lubricating by incorporating polytetrafluoroethylene (PTFE) in a liner in the bore for use in a temperature range of -65° to +325°F.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-S-735	Standard Test Fluids; Hydrocarbon.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-197	Bearings, Anti-Friction, Associated Parts and Sub-Assemblies, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, SESD (Code 53), Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS (Continued)

MILITARY (Continued)

MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance.
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number O-148.
MIL-A-8243	Anti-Icing and Deicing-Defrosting Fluid.
MIL-B-81820	Bearing, Plain, Self-Aligning, Self-Lubricating, Low Speed Oscillation.
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydro- carbon Base, Aircraft, NATO Code Number H-537.
MIL-B-81934/1	Bearing, Sleeve, Plain, Self-Lubricating, +325°F.
MIL-B-81934/2	Bearing, Sleeve, Flanged, Self-Lubricating, +325°F.

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices.
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-1599	Bearings, Control System Components and Associated Hardware Used in the Design and Construction of Aerospace Mechanical Systems and Subsystems.
MS14101	Bearing, Plain, Self-Lubricating, Self-Aligning, Low Speed, Narrow, Grooved Outer Ring, -65°F to +325°F.

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American National Standards Institute

ANSI B46.1	Surface Texture, Surface Roughness, Waviness and Lay.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

American Society for Testing Materials

F-25	Method for Sizing and Counting Airborne Particulate Contamination in Clean Rooms and Other Dust Controlled Areas Designed for Electronic and Similar Applications.
F-50	Practice for Continuous Sizing and Counting of Airborne Particles in Dust Controlled Areas Using Instruments Based Upon Light Scattering Principles.
C-794	Test Method for Adhesion-in-Peel of Elastomeric Joint Sealants.

(Copies of the above publications may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103, telephone 215-299-5400.)

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Bearings furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.3, 6.3 and 6.3.1).

3.1.1 Product design change. Any change in product design, description, materials or processing procedures will require requalification of the product to an extent determined by the qualifying activity.

3.1.2 Product manufacture. Except for the bonding operations, the manufacturer is permitted to subcontract manufacturing operations without violating the requirements of 3.1.1. The bonding operations shall be performed in the plant listed on the Qualified Products List. Manufacture of the self-lubricating liner material may also be subcontracted. Any change in (1) the liner manufacturer, (2) the liner manufacturing procedures, or (3) the materials used in manufacture of the liner will require requalification to an extent determined by the qualifying activity.

3.2 Materials. Material for the sleeve and liner shall be in accordance with the applicable military specification sheet. PTFE shall be included in the liner in such a manner that the bearing will conform to all requirements of this specification.

3.3 Design. Bearing design shall conform to that shown on MIL-B-81934/1 and MIL-B-81934/2.

3.4 Construction. The liner shall be so secured that all relative motion will be between the liner and the shaft. Except as otherwise specified on the applicable military specification sheet, the details of the design shall be optional.

3.4.1 Dimensions and tolerances. Dimensions and tolerances shall be as specified on the applicable military specification sheet. Dimensions not shown shall be at the option of the manufacturer.

3.4.2 Surface texture. The surface texture shall be in accordance with the applicable specification sheet and ANSI B46.1. Bearings shall be free of any surface defects which may be detrimental to satisfactory installation, performance or bearing life as defined in this specification.

3.4.3 Lubrication. Initial grease or oil lubrication by the manufacturer will not be permitted.

3.4.4 PTFE liner condition and bond integrity.

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3.4.4.1 Visual examination. The visual appearance of the bonded liner shall exhibit a degree of workmanship consistent with proper manufacturing process controls, as checked per 4.6.5.1. The liner shall be uniform in texture and shall contain no imbedded contaminants. The liner set-back shall meet the applicable drawing requirements. There shall be no separation or lifting of the liner at any of the edges. There shall be no unraveling or excessive fraying of the liner at any of the edges. Molded liners shall not contain embedded contaminants, cracks or bubbles.

3.4.4.2 Bond integrity and peel strength. When tested in accordance with 4.6.5.2, the liner shall be tightly adherent to the metallic substrate over at least 90 percent of the contact area and shall exhibit an average peel strength of 2.0 pounds per inch or greater. The adhesive remaining on the metal substrate shall have no void or unbonded area which cannot be included within a circumscribing circle with a diameter equal to 25 percent of the race width or 0.25 inch, whichever is smaller.

3.4.4.2.1 Processing controls. All fabrication of the PTFE liner involving application or mixing of adhesive, and all liner bonding procedures involving application of adhesive, shall be conducted in a controlled area.

3.4.4.2.1.1 Controlled area. The controlled area shall be maintained at a temperature of 75° ±10°F with a maximum relative humidity of 75%. The enclosed atmosphere of the work area shall be well ventilated and maintained so that the particle count is 2500 maximum 5.0 microns or larger when measured per ASTM F-25 or ASTM F-50. The particle count measurement shall be performed annually (minimum). There shall be no eating or smoking in the controlled area and no process which produces uncontrolled spray, dust, fumes or particulate matter.

3.4.4.3 Liner peelability. Each manufacturer shall establish during qualification testing whether the liner is peelable or non-peelable. If peelable, the manufacturer shall determine the mean and standard deviation peel strength values for the liner based upon a minimum of six peel strength tests conducted in accordance with 4.6.5.2. This data shall be recorded in the qualification test report. A liner originally qualified as peelable shall remain peelable in production. A liner originally qualified as non-peelable shall remain non-peelable in production.

3.5 Performance.

3.5.1 Radial static limit load. After the radial static limit load listed in Table I has been applied as specified in 4.6.1, the permanent set shall not exceed .002.

3.5.2 Oscillation under radial load. The total liner wear of the bearing shall not exceed 0.0035 inch after 1000 cycles, 0.0040 inch after 5000 cycles and 0.0045 inch after 25,000 cycles when tested at room temperature in accordance with 4.6.2. Visual examination of the liner after test shall indicate no loss of bonding to the metal substrate in the loaded area.

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TABLE I. Load values.

Part no.	Radial static limit load (lbs)	Oscillation load (lbs)	After test torque (in./lbs max)
M81934/1-08A012	6,900	6,300	79
-08C012	10,800	6,300	79
-16A016	20,000	16,500	410
-16C016	31,400	16,500	410
-24A016	30,000	22,500	840
-24C016	47,100	22,500	840

3.5.3 Fluid compatibility. When tested in accordance with 4.6.3, the bearings shall be compatible with the fluids listed in 4.6.3 and the total liner wear shall not exceed 0.0060 inch. Visual examination of the liner after test shall indicate no loss of bonding to the metal substrate in the loaded area.

3.5.4 High temperature. When tested in accordance with 4.6.4, under the dynamic load specified in Table I, the total liner wear shall not exceed 0.0060 inch. Visual examination of the liner after test shall indicate no loss of bonding to the metal substrate in the loaded area.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other and with respect to installation and performance. The drawing number requirements of DOD-STD-100 shall govern documentation of and changes in the manufacturer's part numbers.

3.7 Identification of product. Each bearing shall be permanently and legibly marked with the manufacturer's identification. Where space permits, other information as specified on military specification sheet shall be marked. Metal impression stamping is prohibited.

3.8 Workmanship.

3.8.1 General requirements. All workmanship shall be such as to result in a first quality bearing which meets the requirements of this specification. The bearing shall be free from defects which may affect its durability and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Qualification test records. The manufacturer shall maintain a record showing quantitative results for all tests required by this specification. This record shall be available to the purchaser and shall be signed by an authorized representative of the manufacturer or the testing laboratory, as applicable.

4.2 Classification of tests. The inspection and testing of the bearings shall be classified as:

a. Qualification tests (4.3)

b. Quality conformance tests (4.4)

4.3 Qualification tests.

4.3.1 Sampling instructions. Qualification test samples shall consist of 38 bearings conforming to M81934/1-08A012 plus 15 bearings of each of the additional bore diameters, widths and housing materials specified below for which qualification is desired. All bearings necessary for tests specified herein shall be furnished by the manufacturer. Samples shall be identified as required and forwarded to the activity designated in the letter of authorization (see 6.3 and 6.3.1).

When approved:
M81934/1-08C012 will qualify M81934/1&/2-04C thru -09C
M81934/1-08A012 will qualify M81934/1&/2-04A thru -09A
M81934/1-16C016 will qualify M81934/1&/2-10C thru -18C
M81934/1-16A016 will qualify M81934/1&/2-10A thru -18A
M81934/1-24C016 will qualify M81934/1&/2-20C thru -32C
M81934/1-24A016 will qualify M81934/1&/2-20A thru -32A

4.3.2 Certified test report. The manufacturer shall furnish a certified test report showing that the manufacturer's product satisfactorily conforms to this specification (see 6.3.1). The test report shall include, as a minimum, actual results of the tests specified herein. When the report is submitted, it shall be accompanied by a dated drawing which completely describes the manufacturer's product by specifying all dimensions and tolerances, outer race material, coating or plating, and heat treatment. The manufacturer's part number for each size shall be included on the drawing.

4.3.3 Tests. Qualification tests shall include all the examinations and tests of this specification. The minimum number of samples per test shall be in accordance with Table II.

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4.3.4 Retention of qualification. The continued listing of a product on the Qualified Products List is dependent upon a periodic verification of the manufacturer's continued compliance with the requirements of this specification and with standardization regulations. As part of that verification process, each manufacturer must complete DD Form 1718 during October of each odd numbered year. This form, supplied by the qualifying activity, is to be signed by a responsible official of management and sent to the Naval Air Engineering Center, SESD Code 5311, Lakehurst, NJ 08733-5100.

4.4 Quality conformance tests. The quality conformance tests of the bearings shall consist of the examinations and tests of Table III to determine conformance of the bearings to the requirements of this specification. Unless otherwise specified by the purchaser, inspections shall be conducted in accordance with MIL-STD-105, Table II-A. In addition, when so required by the purchase order, supplemental tests shall be performed to determine conformance to the requirements of this specification not covered in Table III.

TABLE II. Qualification test samples.

Examination and tests	Paragraph number	Samples to be tested
Examination of product	4.5.1	5
Preparation for delivery	4.5.2	5
*Radial static limit load	4.6.1	3
*Oscillation under radial load	4.6.2	3
*Fluid compatibility	4.6.3	3 (M81934/1-08A012 only)
*High temperature	4.6.4	3
Bond integrity	4.6.5	1

*These tests are not required if:

1. The bearing manufacturer is qualified to MIL-B-81820 for sizes MS 14101-5, -8 and -12.

2. The liner materials and bonding procedures used in these sleeve bearings are the same as qualified to MIL-B-81820.

TABLE III. Quality conformance tests.

Examinations and tests	Paragraph numbers	AQL
(a) Dimensions	(3.4.1), (4.5.1)	4.0
(b) Identification of product	(3.7), (4.5.1)	1.0
(c) Workmanship	(3.4.4.1), (3.8), (4.5.1)	1.0
(d) Preparation for delivery	(4.5.2)	1.0
(e) Liner condition and bond integrity	(3.4.4.1), (3.4.4.2), (4.6.5)	10.0

4.4.1 Inspection lot. The inspection lot shall consist of finished bearings, having a single part number, manufactured according to same procedures as the parts originally qualified and produced as one continuous run or order or portion thereof.

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4.4.2 Sampling.

4.4.2.1 Sample for quality conformance tests (a) through (d). The sample bearings shall be selected from each inspection lot in accordance with MIL-STD-105, inspection level II.

4.4.2.2 Sample for quality conformance test (e). The sample bearing shall be selected from each inspection lot in accordance with MIL-STD-105, inspection level S-2.

4.4.3 Quality assurance certification. For each inspection lot the manufacturer shall maintain and supply to the purchaser upon demand:

a. Certified copies of all records of quality conformance tests specified in 4.4 and the purchase order.

b. Certification that the materials, manufacturing procedures and processes used in producing the bearings are the same as those of the bearings originally qualified.

These records and certifications shall identify the manufacturer of the bearings, the address of the plant where they were manufactured, the purchaser and the purchase order number.

4.4.4 Resubmitted inspection lots. The paragraph titled "Resubmitted lots or batches" of MIL-STD-105 shall apply. A resubmitted inspection lot shall be inspected using tightened inspection. Where the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be selected. When an inspection lot is resubmitted, full particulars concerning the cause of previous rejection and the action taken to correct the defects found in the inspection lot, shall be furnished by the contractor to the procuring activity.

4.5 Examinations.

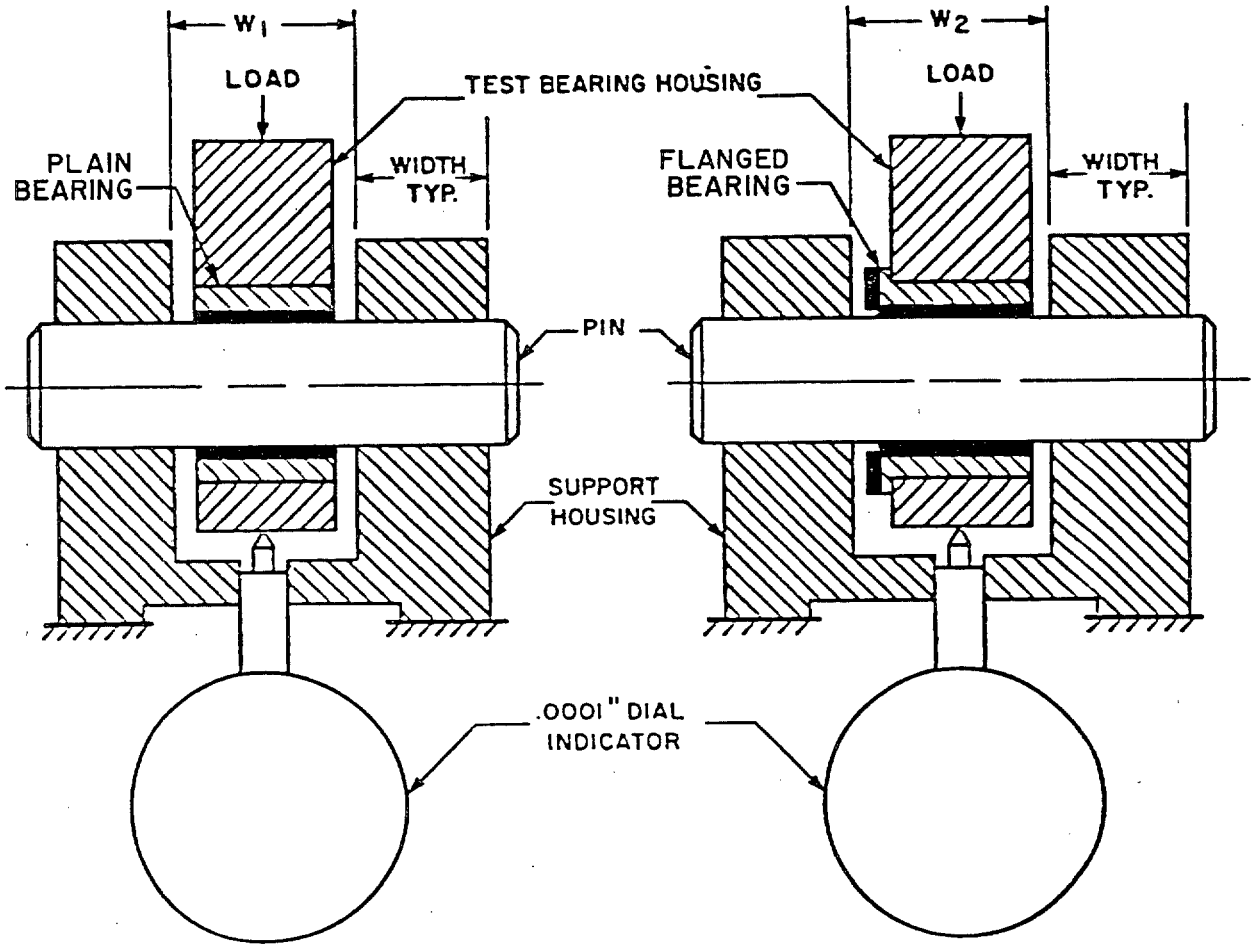
4.5.1 Examination of product. The bearings shall be examined to determine conformance to the requirements of this specification and the applicable military specification sheet for material, dimensions, finish, identification of product, workmanship and requirements not covered by tests.

4.5.2 Preparation for delivery. Preservation, packaging, packing and marking shall be inspected to determine conformance to Section 5.

4.6 Test methods. Unless otherwise specified, all tests shall be conducted at room temperature.

4.6.1 Radial static limit load. The bearings shall be installed in a test fixture as shown on Figure 1, using .0001 to .0011 (CRES) and .0001 to .0016 (aluminum) interference fit with the housing and a 0.002 to 0.004-inch loose fit with the pin. A preload of 4 to 6 percent of the radial static limit load shall be applied to the bearing for 3 minutes, and the measuring device set at zero. The load shall then be increased at the rate of 1 percent

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Pin: Diameter = B -.000/-.001
Material = Steel 50 Rc min
Finish = 8 " max and shall be honed, polished or similarly finished subsequent to grinding
Note: Taper one end .5" min length, tapered end to extend beyond support housing when assembled

Test Bearing Housing: Bore = D -(.0006 to .0011)
Width = L max + .002 -.000
Length = 2D min

Support Housing: Bore = Pin dia max + (.0002 to .0027)
Min support width = L nominal
Length = (2) (pin dia min)
W_1 = L + (.025 to .030)
W_2 = L + F + (.025 to .030)

(L, F and D may be obtained from the appropriate specification sheet)

FIGURE 1. Radial test fixture.

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of the specified load per second until it equals the radial static limit load listed in Table I. The load shall then be reduced at the same rate to the preload value. The permanent set shall be the reading at preload.

4.6.2 Oscillation under radial load. The bearing shall be installed in a steel housing, using a .0001 to .0011 (CRES) and .0001 to .0016 (aluminum) inch interference fit with the housing and a 0.000 to 0.001-inch loose fit with the pin. The bearing shall be so installed as to place the pin in double shear. A dial indicator or electronic pickup shall be so mounted that any radial movement of the pin or the bore of the bearing with respect to the bearing outside diameter can be measured. The oscillation load specified in Table I shall be applied and held statically for 15 minutes. At the end of this time, the indicating device shall be set at zero and the oscillating test shall be started. Wear readings shall include the wear from the first cycle on. The test shall be run in such manner that the pin is oscillated ±25 degrees (50 degrees total) at 10 cycles per minute for 25,000 cycles. One cycle shall consist of rotation from zero degrees to +25 degrees, return through zero degrees to -25 degrees and return to zero degrees. Sufficient readings during the test shall be recorded to plot a graph of wear (thousandths of an inch) vs life (cycles). Upon completion of the test, the loaded breakaway torque shall be as specified in Table I and liner wear and liner bond shall be as specified in 3.5.2.

4.6.3 Fluid compatibility. Fifteen bearings conforming to M81934/1-08A012 (3 for each fluid) shall be immersed for 24 hours in each of the following fluids at 160° ±5°F, except for (b) which shall be at 110° ±5°F:

a. Skydrol 500B hydraulic fluid

b. TT-S-735, type VII standard test fluid or MIL-T-5624, grade JP-4 or JP-5

c. MIL-L-7808 lubricating oil

d. MIL-H-5606 hydraulic oil

e. MIL-A-8243 anti-icing fluid

f. MIL-H-83282 hydraulic fluid

Within 1/2 hour after removal from the test fluid the bearing shall be tested in accordance with 3.5.3 and 4.6.2. The load for (e) shall be 75 percent of the oscillation load specified in Table I.

4.6.4 High temperature. Three bearings conforming to M81934/1-08A012 shall be subjected to the test of 4.6.2, except that the bearings shall be heated in such a way that the pin/liner interface is maintained at a temperature of 325°F minimum. The load shall be 100% of the oscillation load specified in Table I.

4.6.5 Bond integrity.

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4.6.5.1 Visual examination. Visually examine the exposed liner surface for conformance to the applicable requirements of 3.4.4.1, 3.8 and the military specification sheet.

4.6.5.2 Bond integrity and peel strength. Peel the liner away from the metal substrate and evaluate the peel strength and adhesive bond appearance per 3.4.4.2. A blade or scribe may be used to initiate the peel. To determine peel strength, the liner shall be attached to a calibrated spring scale or tension testing machine of the type described in ASTM C-794 and the liner peeled back from the substrate at 140° ±40° angle to the bond surface at a speed of 0.5 to 1.0 inch per minute (see Figure 2). The average peel strength value shall be recorded and included in the test report. Where possible, the peel shall be conducted on the entire width of the bonded liner. When the peel is conducted on the entire width of the race, the width for calculating peel strength in pounds per inch shall be the machined width minus 0.050 inch to account for chamfers and edge effects. At the manufacturer's option, or when peeling of the full race width is impractical, the liner may be cut through to the metal substrate to form a peel sample with a minimum width of 0.500 inch. This peel sample may be cut either parallel to or perpendicular to the race side face. This sample shall be peeled as previously described. The peel strength test shall be waived for the -04 through -09 size bearings, but the liner shall be peeled by hand to permit visual examination of the bond line for voids.

Following liner peel, the adhesive bond appearance shall be evaluated to determine the location and size of any voids or unbonded areas as specified in 3.4.4.2. In the event the liner cannot be removed without employing chipping, scraping or abrasive techniques, the liner shall be considered to be properly bonded and free of voids and shall be classed as nonpeelable.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or C as specified in 5.1.1 and 5.1.2 (see 6.2). Cleaning solvent, grease, or oil shall not be used. Prior to packaging, bearings shall be cleaned in one of the following ways:

a. Clean, dry compressed air per MIL-P-116.

b. Wiping with soft, clean cloth per MIL-P-116.

c. Manufacturer's established cleaning procedures for this bearing.

5.1.1 Level A. Packaging shall be in accordance with Level A of MIL-B-197. The manufacturer's lot control number shall be marked on each package.

5.1.2 Level C. Packaging shall be in accordance with the fabricator's commercial practice. The manufacturer's lot control number shall be marked on each package.

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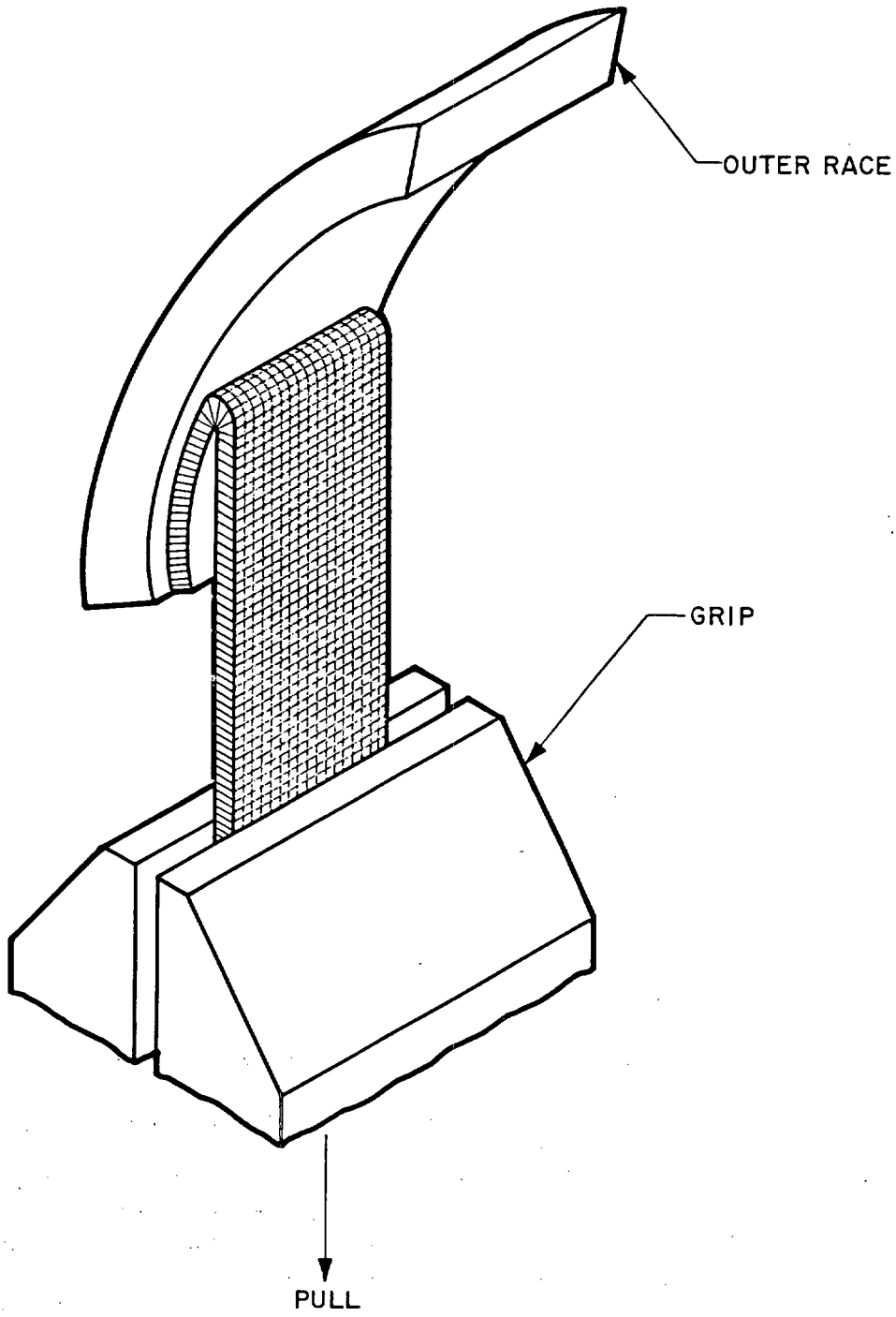


FIGURE 2. Example of peel test fixture.

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5.2 Packing.

5.2.1 Level A. Bearings packaged as specified in 5.1.1 shall be packed for overseas shipment and storage in accordance with MIL-B-197.

5.2.2 Level B. Bearings packaged as specified in 5.1.1 shall be packed for domestic shipment and storage in accordance with MIL-B-197.

5.2.3 Level C. Bearings packaged as specified in 5.1.2 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. The containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers, as applicable to the mode of transportation.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall include:

Bearings, Sleeve, *(Plain or Flanged), Self-Lubricating

*Applicable data to be entered by contractor.

6. NOTES

6.1 Intended use. These bearings are intended primarily for use in airframe applications of high loads at low rotational oscillatory speeds. For specific design information on the capability of these bearings under particular load speed and wear/life conditions the user is referred to MIL-STD-1599.

6.2 Ordering data. Procurement documents should specify:

a. Title, number and date of this specification.

b. Military identifying part number.

c. Applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

d. Quality assurance certification (see 4.4.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List 81934 whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command (Naval Air Engineering Center, Systems Engineering and Standardization Department, Code 5311, Lakehurst, NJ 08733-5100) and information pertaining to qualification of products may be obtained from that activity; however, information pertaining to qualification of products may be obtained from the Naval Air Development Center, Code 6061, Warminster, PA 18974-5000.

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6.3.1 Authorization for submittal of samples. A manufacturer seeking qualification approval of his product will be authorized to submit samples for such approval only upon presentation of certified test reports and drawings indicating that his product conforms to this specification.

6.4 Definitions.

6.4.1 Processing procedures. All bonding, curing and postcuring procedures (see 3.1.1).

6.4.2 Void. A void is an area where the metal substrate is smooth and shiny with no visible adhesvie (see 4.6.5.2).

6.4.3 Unbonded area. An unbonded area is where the adhesive remaining on the metal substrate is smooth and shiny indicating a lack of bonding pressure.

6.5 Changes from previous issue. The margins of this standard are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.6 Subject term (key word) listing.

Bearings, self-lubricating
Bearings, sleeve

Custodians:
Navy - AS
Air Force - 11
Army - AV

Reviewer activities:
Navy - SH
Air Force - 99, 84
Army - AR, AT
DLA - IS

Preparing activity:
Navy - AS
(DOD Project No. 3120-0663)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-81934A	SPECIFICATION FOR BEARINGS, SLEEVE, PLAIN AND FLANGED, SELF-LUBRICATING, GENERAL

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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